General Urine Examination

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First stage

Lab 7

General Urine Examination or. Urinalysis

Why urinalysis?

- General evaluation of health
- Diagnosis of disease or disorders of the kidneys or urinary tract
- Diagnosis of other systemic disease that affect kidney function
- Monitoring of patients with diabetes
- Screening for drug abuse (eg. Sulfonamide or aminoglycosides)

General Urine Examination

Urinalysis consists of the following measurements:

- Macroscopic or physical examination
- Chemical examination
- Microscopic examination of the sediment
- Urine culture

Physical examination of urine

- Examination of physical characteristics:
- Volume
- Color
- Odor
- pH
- Specific gravity

Physical examination (Volume)

- Normal- 1-2.5 L/day
- Anuria- Urine output $<$ 100ml/day
- Oliguria- Urine Output $<$ 400ml /day
- Polyuria- Urine Output $>$ 2.5 L/day

Physical examination (color)

- Normal color: pale yellow (amber yellow)due to the presence of pigments of urobilin or urobilinogen
- Abnormal colors of urine:

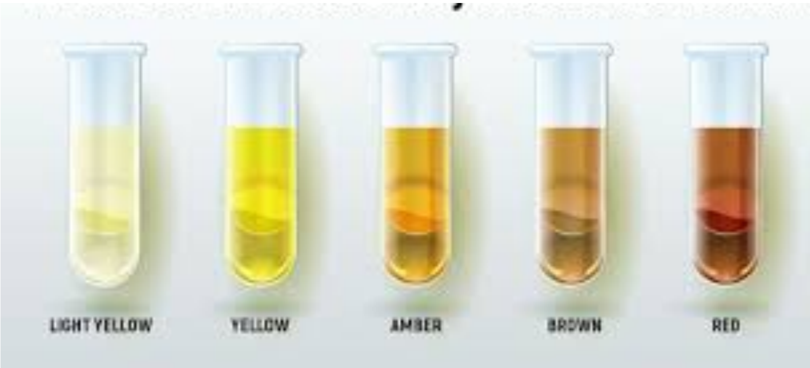
➢ Colorless: Diabetes

➢ Orange :Ingestion of large amount of carotenoids (vitamin A)

➢ - Yellowish brown :Hepatic Jaundice

➢ Black : Melanin (melanoma)

➢ Smoky :presence RBCs



Physical examination (Odor)

- Normal - aromatic due to the volatile fatty acids
- On long standing – ammonical (decomposition of urea forming ammonia which gives a strong ammonical smell)
- Foul, offensive - pus or inflammation
- Sweet - Diabetes
- Fruity - Ketonuria

Physical examination (pH)

- Urine pH ranges from 4.5 to 8
- Normally it is slightly acidic lying between 6 – 6.5, Tested by:

➢ litmus paper

➢ pH paper

➢ dipsticks

- Acidic Urine –Ketosis (diabetes, starvation, fever),systemic acidosis, UTI- E.coli

Physical examination (Specific gravity)

- It is measurement of urine density which reflects the ability of the kidney to concentrate or dilute the urine relative to the plasma from which it is filtered.
- Measured by:
- Urinometer
- Refractometer
- Dipsticks

Microscopic examination of urine

- A sample of well-mixed urine (usually 10-15 ml) is centrifuged in a test tube at relatively low speed (about 2000-3,000 rpm) for 5-10 minutes which produces a concentration of sediment (cellular matter) at the bottom of the tube.

- A drop of sediment is poured onto a glass slide, a thin slice of glass (a coverslip) is place over it and observed under microscope

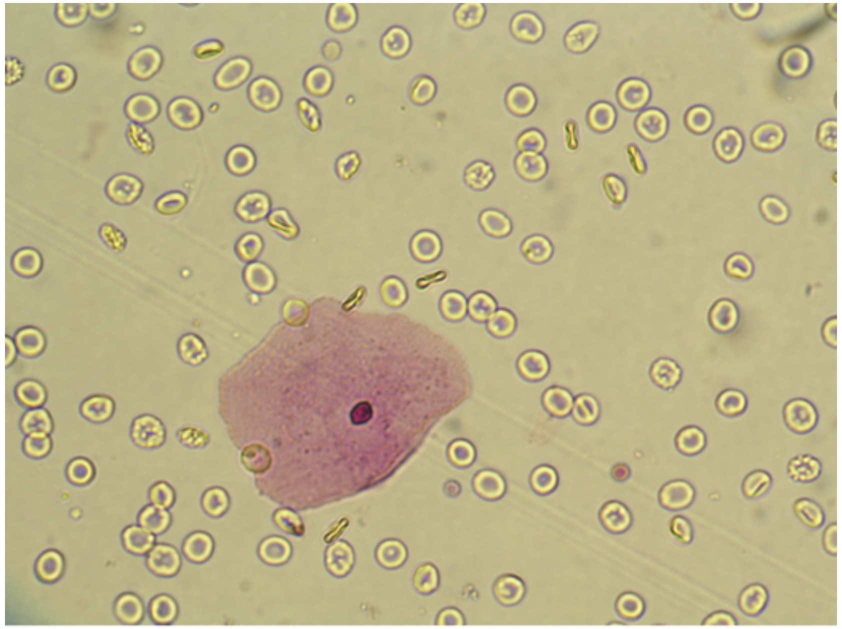
Microscopic examination of urine

- A variety of normal and abnormal cellular elements may be seen in urine sediment such as:
- Red blood cells
- White blood cells
- Mucus
- Various epithelial cells
- Various crystals
- Bacteria
- Casts
- Yeast
- Parasite

Microscopic examination of urine •

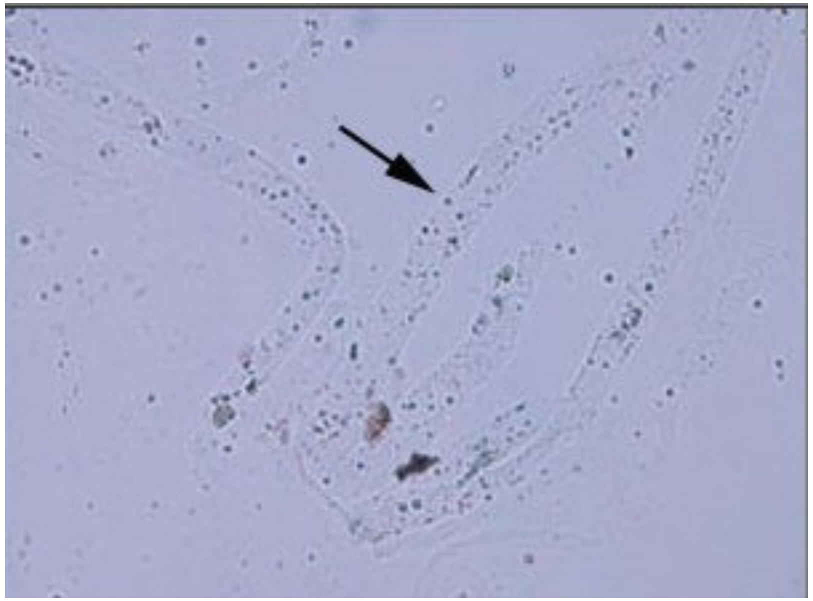
- WBC in high numbers indicate inflammation or infection somewhere along the urinary

RBC indicate infection, trauma, kidney inflammation (glomerulonephritis) ,kidney stones or tumors



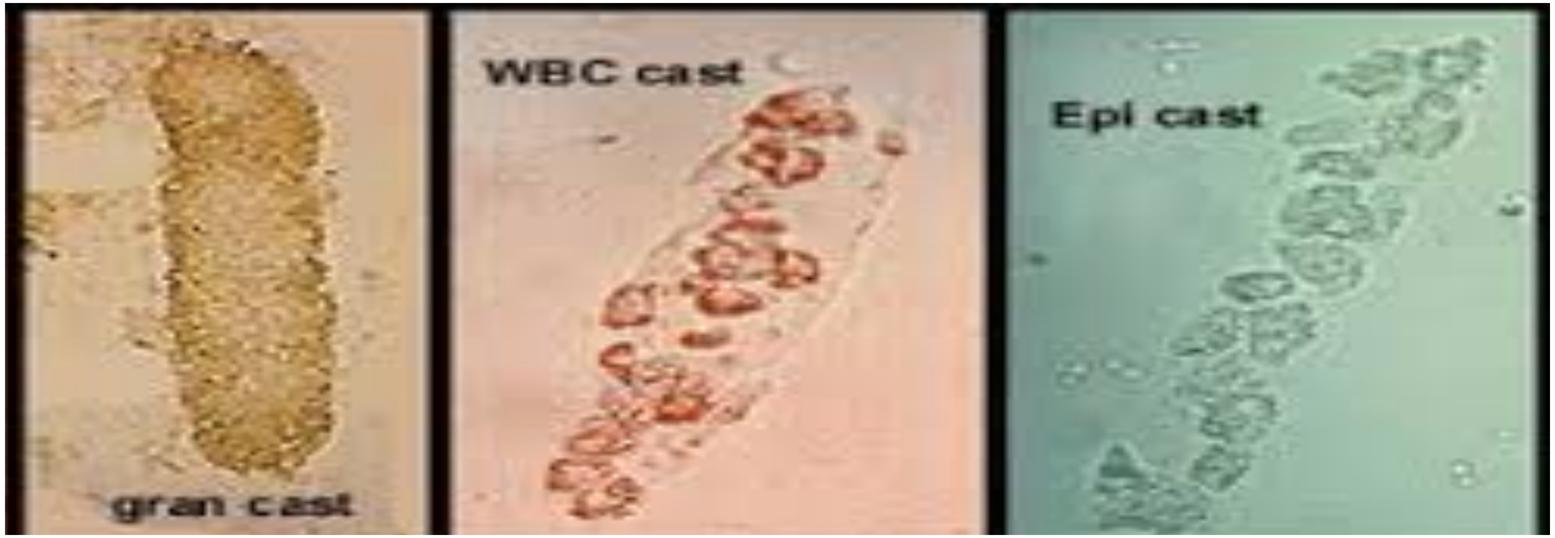
Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition:* www.accessmedicine.com

Mucus threads in urine :An excess amount may indicate a urinary tract infection (UTI)



Microscopic examination of urine Casts

- Urinary casts are cylindrical aggregations of particles that form in the distal nephron, dislodge, and pass into the urine. In urinalysis they indicate kidney disease.



Microscopic examination of urine A variety of normal and abnormal crystals may be present in the urine sediment

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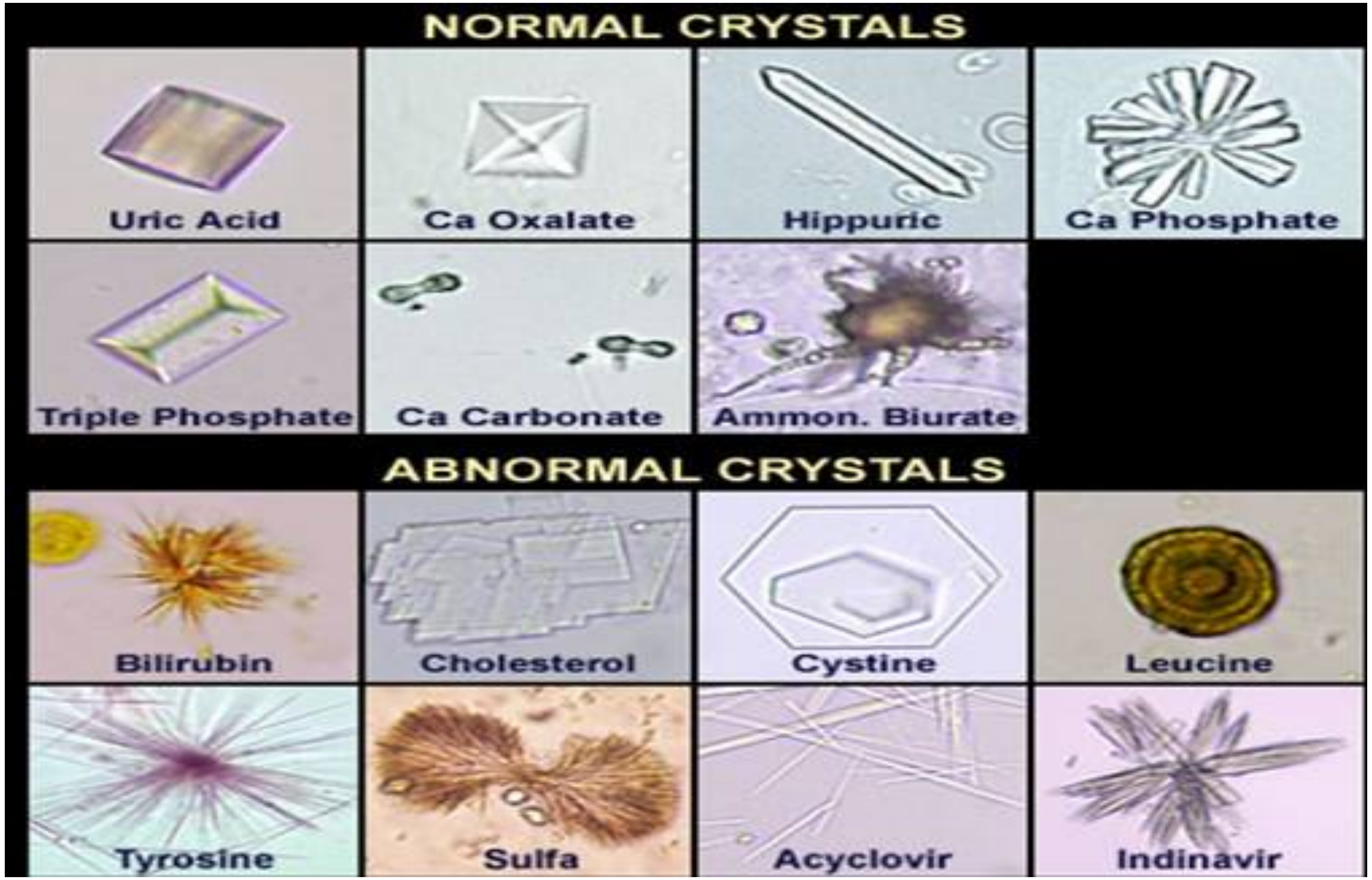


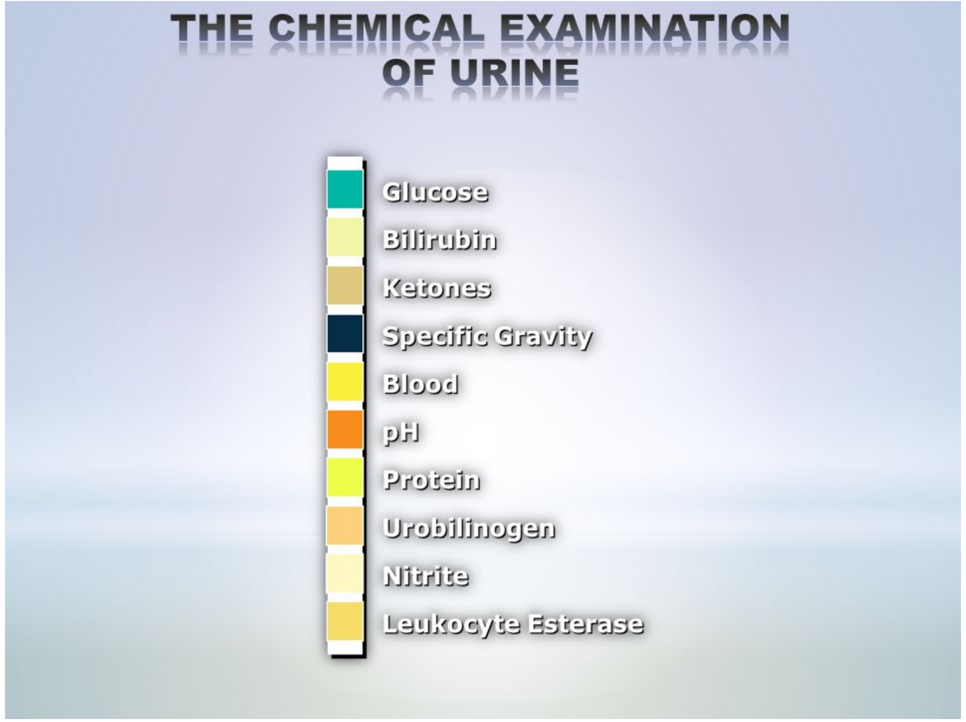
Chemical analysis of urine

- The chemical analysis of urine us undertaken to evaluate the levels of the following component: –
- Protein
- Glucose
- Ketones
- Occult blood
- Bilirubin
- Urobilinogen
- Bile salts

Chemical analysis of urine

The presence of normal and abnormal chemical elements in the urine are detected using dry reagent strips called dipsticks.

- When the test strip is dipped in urine the reagents are activated and a chemical reaction occurs.
- The chemical reaction results in a specific color change.
- After a specific amount of time has elapse, this color change is compared against a strip


THE CHEMICAL EXAMINATION
OF URINE
Glucose
Bilirubin
Ketones
Specific Gravity
Blood
pH
Protein
Urobilinogen
Nitrite
Leukocyte Esterase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